VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2018

H2 Mathematics 9758/02

Paper 2 3 hours

Additional Materials: Answer Paper
Graph Paper (Upon Request)
List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
You are expected to use an approved graphing calculator.
Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.
Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.
You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Section A: Pure Mathematics [40 marks]

1 The function f is defined by

$$\mathrm{f}: x \mapsto \lambda + \frac{1}{1-x}, \quad x \in \mathbb{R},\ x \neq 1,$$

where λ is a negative constant.

(i) Show that f^{-1} exists. [2]

(ii) Find the set of values of λ such that the equation $\mathrm{f}(x) = 2x$ has real solutions. [3]

The functions g and h are defined by

$$\mathrm{g}: x \mapsto \mathrm{f}(x),\ x < 0,$$

$$\mathrm{h}: x \mapsto \left(x - \left(\lambda + \frac{2}{3}\right)\right)^2, x \in \mathbb{R}.$$

(iii) Find the range of hg. [3]

2 **(i)** Show that $2\sin\left(\frac{1}{2}\right)\sin(n) = \cos\left(\frac{2n-1}{2}\right) - \cos\left(\frac{2n+1}{2}\right)$. [2]

(ii) By expressing $\sin(n)$ as $\frac{1}{2}\operatorname{cosec}\left(\frac{1}{2}\right)\left(2\sin\left(\frac{1}{2}\right)\sin(n)\right)$, find

$\sin(1) + \sin(2) + \sin(3) + \ldots + \sin(N)$ in terms of N, where $N \in \mathbb{Z}^+$. [3]

(iii) Explain why the above series will not be equal to $\frac{1}{2}\cot\left(\frac{1}{2}\right)$ for all $N \in \mathbb{Z}^+$. [1]

(iv) Use your answer in part **(ii)** to find the numerical value of $\sum_{n=10}^{25} \sin(n)$. [2]

3 It is given that

$$\mathrm{f}(x) = \begin{cases} 2\sqrt{1+x^2} & \text{for } -1 \leq x \leq 1, \\ 1 & \text{for } 1 < x < 3, \end{cases}$$

and that $\mathrm{f}(x+4) = \mathrm{f}(x)$ for all real values of x.

(i) Sketch the graph of $y = \mathrm{f}(x)$ for $-1 \leq x < 7$. [4]

(ii) Using the substitution $x = \tan\theta$, show that $\int_3^5 \mathrm{f}(x)\,\mathrm{d}x = 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^3\theta\,\mathrm{d}\theta$.

By writing $\sec^3\theta$ as $\sec\theta\sec^2\theta$, find the value of $\int_3^5 \mathrm{f}(x)\,\mathrm{d}x$, leaving your answer in the form $a + \ln b - \ln c$, where a, b and c are constants to be determined exactly. [8]

4 Refinement is the process that mined material undergoes so as to recover the desired metals and separate all the undesirable minerals. Recovery rate measures the amount of metal extracted, as a percentage of the actual metal content. There are various refinement methods where the mined material undergoes numerous repetitions of separation reactions to reach the desired recovery rate. We will take the desired recovery rate to be 60% for this question.

(i) The following claim is made by a company using Method *A*:
Method *A* recovers 2% of the metal content after the first reaction. For each subsequent reaction, the amount of metal recovered is an additional t % compared to the amount recovered in the previous reaction. Hence, the amount of metal recovered during the first reaction is 2%, the amount of metal recovered during the second reaction is $(2 + t)$%, the amount of metal recovered during the third reaction is $(2 + 2t)$%, and so on.

Find the value of t such that the total percentage of metals recovered after 20 reactions reaches the desired recovery rate. [2]

Give a reason why the claim made by the company may not be realistic in the long run. [1]

(ii) Method *B* recovers 10% of the initial amount of the desired metals after the first reaction, but each subsequent reaction will recover r times as much as the previous reaction. If 10 reactions are required to reach the same desired recovery rate, show that r satisfies the equation $r^{10} - 6r + 5 = 0$. Explain why r cannot be 1, even though $r = 1$ is a root of this equation. Find the value of r. [4]

Each reaction for method *B* will require chemical *L*. At the start of the refinement process, there is 100 kg of chemical *L* and after each reaction, 30% of *L* will be used. 15 kg of chemical *L* will be added at the start of the next reaction.

(iii) Calculate the amount of chemical *L* left after 3 reactions. [2]

(iv) Express the amount of chemical *L* left after n reactions in the form $k\left(0.7\right)^{n-1}+m$, where k and m are constants to be determined. Hence, find the amount of chemical *L* left at the end of a reaction in the long run. [3]

Section B: Statistics [60 marks]

5 Twelve cards, numbered from 1 to 12 are arranged in a straight line. Find the number of ways this can be done if

(i) there are no restrictions, [1]

(ii) the cards numbered 2 and 3 are together, and all the six even numbered cards are adjacent. [3]

The twelve cards are arranged in a circle. Find the number of ways this can be done if all the cards numbered as multiples of 3 are separated. [2]

Six of the cards are selected at random, without replacement. Find the probability that at least two of the chosen cards are even numbered. [3]

6 A wildlife biologist wishes to test frogs for a genetic trait. He caught 1200 frogs from the wild and randomly packed them into boxes of 6. The number, x, of frogs found with the genetic trait in each box is recorded. The results obtained from 200 boxes are shown in the table below.

Number of frogs in box with genetic trait (x)	0	1	2	3	4	5	6
Number of occurrences	17	53	65	45	18	2	0

(i) Calculate, from the above data, the mean value of x. [1]

(ii) State, in context, two assumptions needed for the number of frogs carrying the genetic traits in a box to be well modelled by a binomial distribution. [2]

(iii) Assuming that X can be modelled by a binomial distribution having the same mean as the one calculated in part **(i)**, state the values for the binomial parameters n and p. [1]

Another 60 frogs are caught from the same place and packed randomly into ten boxes of 6. Find the probability that at least three of the boxes contain exactly 2 frogs with the genetic trait. [3]

7 A machine in a factory produces steel rods. The machine is reset periodically. After resetting, it should produce rods with mean length 27.0 cm. In order to test whether the mean length of rods produced differs significantly from 27.0 cm, a sample of fifty rods is taken and their lengths, x cm, are summarised as follows.

$$\sum(x-27)=12.0, \quad \sum(x-27)^2=36.4.$$

(i) Calculate unbiased estimates of the population mean and variance of the lengths of steel rods. [2]

(ii) Carry out an appropriate test at the 2.5% level of significance, explaining whether there is a need for the population distribution of the lengths of the steel rods to be known. [5]

(iii) Explain, in the context of the question, the meaning of ‘at the 2.5% level of significance’. [1]

Another machine in the factory produces steel rods with mean length l cm and standard deviation 3 cm. In order to carry out a test to determine whether the mean length of rods produced is more than l cm at 5% significance level, a sample of fifty rods is taken and the sample mean is found to be 23.8 cm.

Given that the test concluded the population mean is more than l cm , find the set of possible values of l. [4]

8 The probability of obtaining a ‘6’ on a biased cubical dice is thrice the probability of rolling every other number on the dice.

Show that probability of obtaining a ‘6’ on the biased dice is $\frac{3}{8}$. [1]

This biased dice is put into a bag together with 3 fair cubical dice.

(i) One of the dice is chosen randomly from the bag and rolled once. Find the probability of obtaining a ‘6’. [2]

(ii) One of the dice is chosen randomly and is rolled n times.

(a) Find the probability that ‘6’ is obtained on all the n rolls. [1]

(b) Given that ‘6’ is obtained for all the n rolls, the probability that the biased dice is chosen is more than 0.95. By forming an inequality in terms of n, solve for the least value of n. [3]

(iii) One of the dice is chosen randomly from the bag and rolled once. The dice is then put back into the bag. Another dice is chosen randomly from the bag and rolled once.

For every ‘6’ obtained, the player gains \$2, otherwise the player loses \$$k$. If the game is fair, that is, expected winnings of the player is \$0, show that the value of k is 0.56. [3]

Find the variance of the player’s winnings. [2]

9 AppleC is a fruit farm that produces Cameo apples. It is known that 8.1% of the apples have a mass more than 125 g and 14.7% of the apples have a mass less than 90 g. It is also known that the masses of this batch of Cameo apples follows a normal distribution.

(i) Show that the mean and standard deviation of the masses for this batch of Cameo apples, correct to 3 significant figures, are 105 g and 14.3 g respectively. [4]

The Cameo apples are packed into bags of 8 apples.

(ii) Find the probability that the mass of a randomly chosen bag of Cameo apples is less than 845 g. State the distribution used and its parameters. [3]

A local distributor imports Cameo apples from AppleC. The distributor also imports Jazz apples from another supplier. It is known that the masses of bags of Jazz apples follow a normal distribution with mean 700 g with standard deviation of 35 g. The retail price of Cameo apples and Jazz apples are \$8.50 per kg and \$7.30 per kg respectively.

(iv) Find the probability that the cost of three bags of Jazz apples is more than twice the cost of a bag of Cameo apples by at least \$1. [4]

10 In an agricultural experiment, a certain fertilizer is applied at different rates to ten identical plots of land. Seeds of a type of grass are then sown and several weeks later, the mean height of the grass on each plot is measured. The results are shown in the table.

Rate of application of fertilizer, x g/m^2	10	20	30	40	50	60	70	80	90	100
Mean height of grass, y cm	6.2	11.4	13.2	14.8	15.8	17.0	19.4	19.4	20.6	20.8

(i) Draw the scatter diagram for these values, labelling the axes clearly. [1]

It is thought that the mean height of grass, y cm, can be modelled by one of the formulae

$$y = ax + b \quad \text{or} \quad y = c \ln x + d$$

where a, b, c and d are constants.

(ii) Find, correct to 4 decimal places, the value of the product moment correlation coefficient between

(a) x and y, **(b)** $\ln x$ and y. [2]

(iii) Use your answers to parts **(i)** and **(ii)** to explain which of $y = ax + b$ or $y = c \ln x + d$ is the better model. [2]

It is required to estimate the value of x for which $y = 17.2$.

(iv) Explain why neither the regression line of x on y nor the regression line of $\ln x$ on y should be used. [1]

(v) Find the equation of a suitable regression line and use it to find the required estimate. [3]

[End of Paper]